To eliminate the concern of Reverse gear getting engaged without operating gear selecting knob

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To Eliminate the Concern of Reverse gear getting engaged without operating gear selecting knob Marazzo model."

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	01.06.19	15.11.19
Completion Date	15.11.19	15.11.19

Current Status: **Closed**

Cross Functional Team

SNo	Description	Name	Designation	Dept	Email
1	Team Leader	Taj Shaikh	Manager	MFG	SHAIKH.TAJ@Mahindra.com
2	Member 1	Santosh Jadhav	Deputy Manager	MQA	JADHAV.SANTOSH@Mahindra.com
3	Member 2	Sachin Bhamre	Deputy Manager	MQA	BHAMRE.SACHIN@Mahindra.com
4	Member 3	Nilesh Deore	Asst. Manager	Mfg	DEORE.NILESH@Mahindra.com
5	Member 4	Pravin Patil	Manager	Tool room	Patil.pravin@Mahindra.com
6	Member 5	Chandu Jadhav	Sr associate	MFG	Jadhav.Chandu@Mahindra.com

Step1. Select Subject

Brief Description : To eliminate the concern of wheel hub bolt loose

What : To eliminate the defect of Reverse gear getting engaged without operating gear selecting knob .

Where : RFI complaint reported .

When : Month of JUN-19 .

Why : V1+ severity concern which will directly affects the Customer Safety .

Who : Demerit Audit GNOVAC check man.

How : total 10 cases reported till date in 3 month's.

Step1. Select Subject

Reverse gear getting engaged without operating gear selecting knob

Reverse gear getting engaged without operating gear selecting knob V1+ Severity concern , Which directly affects the customer safety.

2. Justify the choice (reasons)

PROMISE 2022
AUTOMOTIVE SECTOR

We Will Grow

Revenue 75% | Profit 100%

By

'Enhancing lifestyle' in personal segment
'Enriching livelihood' in commercial segment

Being #1 in UV, SCV, EV and #3 in HCV
Doubling volume from overseas markets

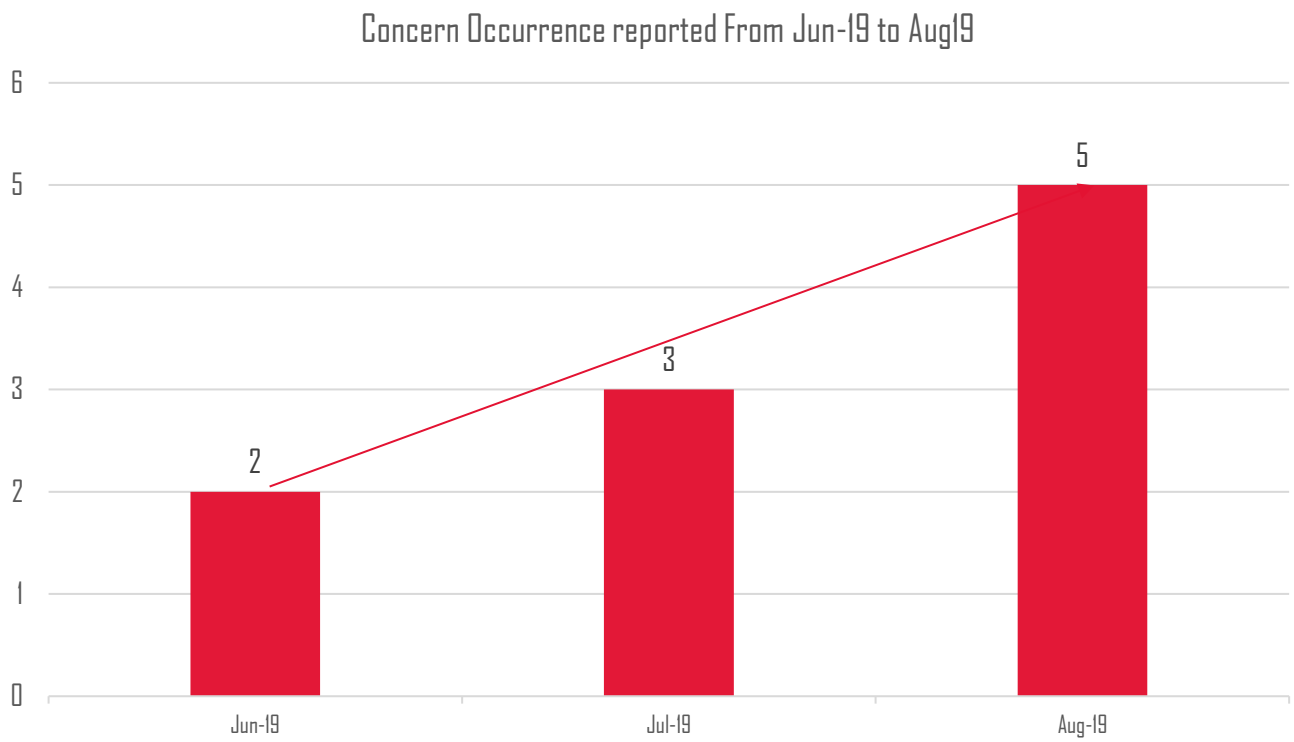
Leading in next generation mobility solutions

#InspiringPeopleToRise

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Inference :

From the above field complaint it is clear Reverse gear getting engaged without operating gear selecting knob V1+ concern

2. Justify the choice

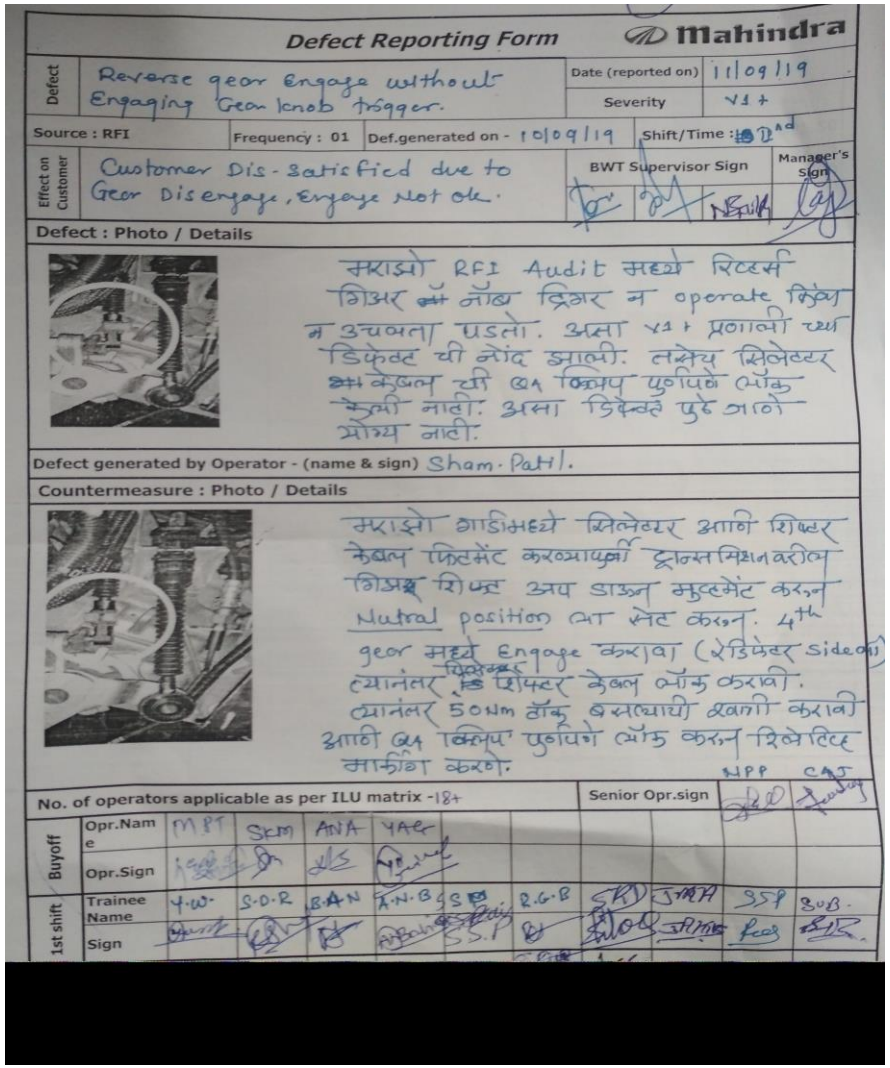
Severity & Consequences of Concern

- Reverse gear getting engaged without operating gear selecting knob is the CC joint & V1+ severity concern.

- If vehicle with Assembly done in 2nd gear will not be engage as per requirement it will directly affect on Customer safety & may lead to accident.

- Due to such issues brand image of Mahindra will also gets violated.

2.B Immediate Containment Action



Defect Reporting Form — Mahindra

Defect	Reverse gear Engage without Engaging Gear knob trigger.	Date (reported on)	11/09/19
		Severity	V1+
Source : RFI	Frequency : 01	Def.generated on - 10/09/19	Shift/Time :
Effect on Customer	Customer Dis-satisfied due to Gear Disengage, Engage Not ok.	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

मराठी RFI Audit मध्ये रिवर्स गिअर नॉब ट्रिगर न operate केल्या न उचलता पडतो. असा V1+ प्रणाली च्या डिफेक्ट ची नोंद झाली. तसेच सिलेक्टर केबल ची QA क्लिप पुर्णपणे लॉक केली नाही. असा डिफेक्ट पुढे जाणे योग्य नाही.

Defect generated by Operator - (name & sign) Sham. Patil.

Countermeasure : Photo / Details

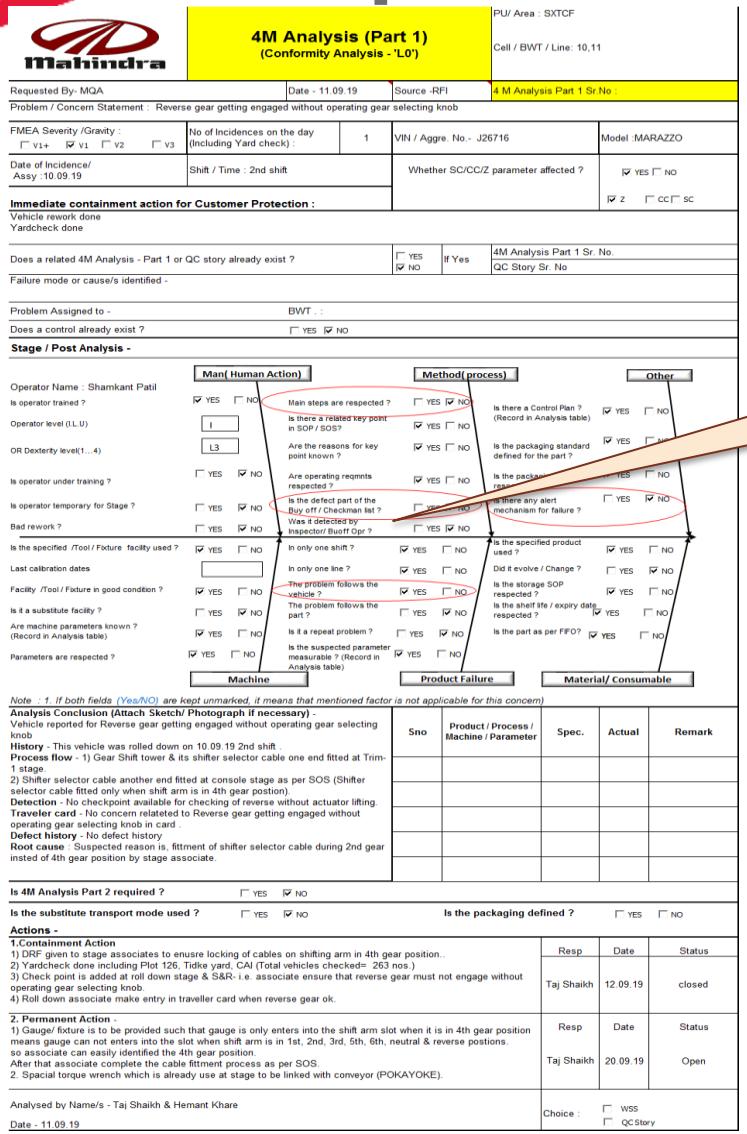
मराठी गाडीमध्ये सिलेक्टर आणि शिफ्टर केबल फिटमेंट करण्यापूर्वी ट्रान्समिशनवरील गिअर शिफ्ट अप डाऊन मुव्हमेंट करून Nutral position ला सेट करून. 4th gear मध्ये Engage करावा (रेडियेटर side ला) त्यानंतर शिफ्टर केबल लॉक करावी. त्यानंतर 50Nm टॉर्क बसवायची खात्री करावी आणि QA क्लिप पुर्णपणे लॉक करून रिलेटिव्ह मार्किंग करणे.

No. of operators applicable as per ILU matrix -18+ | Senior Opr.sign | NPP | CAJ

Buyoff	Opr.Name	MPT	SKM	ANA	YAG						
	Opr.Sign										
1st shift	Trainee Name	Y.W.	S.D.R	B.A.N	A.N.B	S.S.P	R.G.B	SRJ	JMH	SSP	BUB.
	Sign										

1. Awareness Given to All stage, Buy off Associates for the Defect.
2. Yellow marking started on shifter and selector cable.

Step 3. Understanding Current Situation



4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : SXTCF
Cell / BWT / Line: 10,11

Requested By- MQA | Date - 11.09.19 | Source -RFI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Reverse gear getting engaged without operating gear selecting knob

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- J26716 | Model :MARAZZO

Date of Incidence/ Assy :10.09.19 | Shift / Time : 2nd shift | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☑ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
Vehicle rework done
Yardcheck done

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - | BWT . :

Does a control already exist ? ☐ YES ☑ NO

Stage / Post Analysis -

Operator Name : Shamkant Patil

Man(Human Action)		Method(process)		Other	
Is operator trained ?	☑ YES ☐ NO	Main steps are respected ?	☐ YES ☑ NO	Is there a Control Plan ? (Record in Analysis table)	☑ YES ☐ NO
Operator level (I,L,U)	I	Is there a related key point in SOP / SOS?	☑ YES ☐ NO	Is the packaging standard defined for the part ?	☑ YES ☐ NO
OR Dexterity level(1...4)	L3	Are the reasons for key point known ?	☑ YES ☐ NO	Is the packag[illegible]	[illegible] YES ☐ NO
Is operator under training ?	☐ YES ☑ NO	Are operating reqmnts respected ?	☑ YES ☐ NO	Is there any alert mechanism for failure ?	☐ YES ☑ NO
Is operator temporary for Stage ?	☐ YES ☑ NO	Is the defect part of the Buy off / Checkman list ?	[illegible]		
Bad rework ?	☐ YES ☑ NO	Was it detected by Inspector/ Buoff Opr ?	☐ YES ☑ NO		
Is the specified /Tool / Fixture facility used ?	☑ YES ☐ NO	In only one shift ?	☑ YES ☐ NO	Is the specified product used ?	☑ YES ☐ NO
Last calibration dates		In only one line ?	☑ YES ☐ NO	Did it evolve / Change ?	☐ YES ☑ NO
Facility /Tool / Fixture in good condition ?	☑ YES ☐ NO	The problem follows the vehicle ?	☑ YES ☐ NO	Is the storage SOP respected ?	☑ YES ☐ NO
Is it a substitute facility ?	☐ YES ☑ NO	The problem follows the part ?	☐ YES ☑ NO	Is the shelf life / expiry date respected ?	☑ YES ☐ NO
Are machine parameters known ? (Record in Analysis table)	☑ YES ☐ NO	Is it a repeat problem ?	☐ YES ☑ NO	Is the part as per FIFO?	☑ YES ☐ NO
Parameters are respected ?	☑ YES ☐ NO	Is the suspected parameter measurable ? (Record in Analysis table)	☑ YES ☐ NO		
Machine		Product Failure		Material/ Consumable	

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
Vehicle reported for Reverse gear getting engaged without operating gear selecting knob
History - This vehicle was rolled down on 10.09.19 2nd shift .
Process flow - 1) Gear Shift tower & its shifter selector cable one end fitted at Trim-1 stage.
2) Shifter selector cable another end fitted at console stage as per SOS (Shifter selector cable fitted only when shift arm is in 4th gear postion).
Detection - No checkpoint available for checking of reverse without actuator lifting.
Traveler card - No concern relateted to Reverse gear getting engaged without operating gear selecting knob in card .
Defect history - No defect history
Root cause : Suspected reason is, fittment of shifter selector cable during 2nd gear insted of 4th gear position by stage associate.

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action
1) DRF given to stage associates to enusre locking of cables on shifting arm in 4th gear position..
2) Yardcheck done including Plot 126, Tidke yard, CAI (Total vehicles checked= 263 nos.)
3) Check point is added at roll down stage & S&R- i.e. associate ensure that reverse gear must not engage without operating gear selecting knob.
4) Roll down associate make entry in traveller card when reverse gear ok.

Resp	Date	Status
Taj Shaikh	12.09.19	closed

2. Permanent Action -
1) Gauge/ fixture is to be provided such that gauge is only enters into the shift arm slot when it is in 4th gear position means gauge can not enters into the slot when shift arm is in 1st, 2nd, 3rd, 5th, 6th, neutral & reverse postions. so associate can easily identified the 4th gear position.
After that associate complete the cable fittment process as per SOS.
2. Spacial torque wrench which is already use at stage to be linked with conveyor (POKAYOKE).

Resp	Date	Status
Taj Shaikh	20.09.19	Open

Analysed by Name/s - Taj Shaikh & Hemant Khare
Date - 11.09.19

Choice : ☐ WSS ☐ QC Story

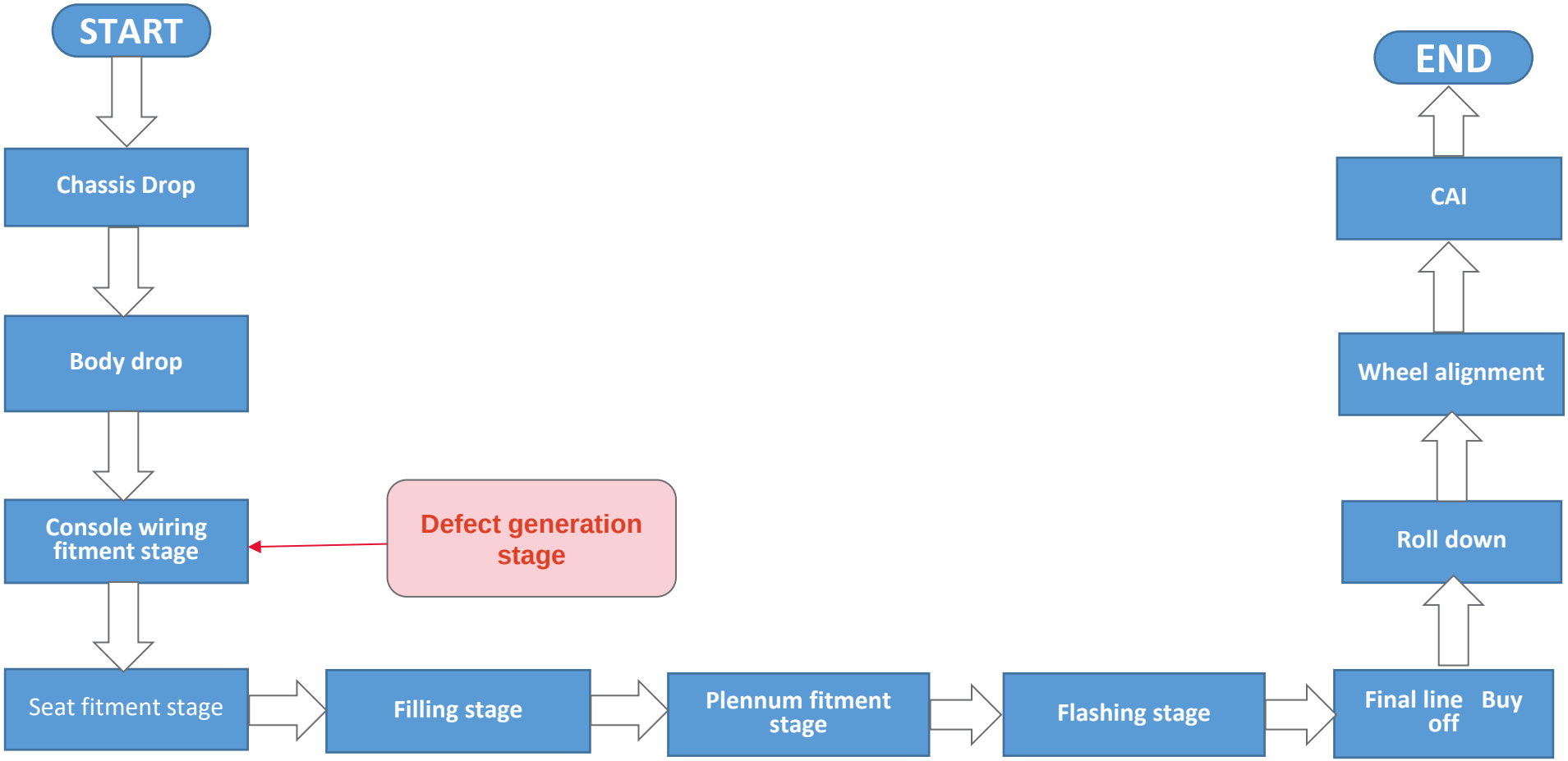
Inference from L0,
Is no detection mechanism for failure and
Fitment done by associate in 2nd gear instead of 4th gear

4M analysis Good and Not so Good observations

MAN	Associate does not follow the main step.
METHOD	No detection mechanism for failure .
OTHERS	No alert mechanism for failure
PRODUCT	Problem follows the Vehicle

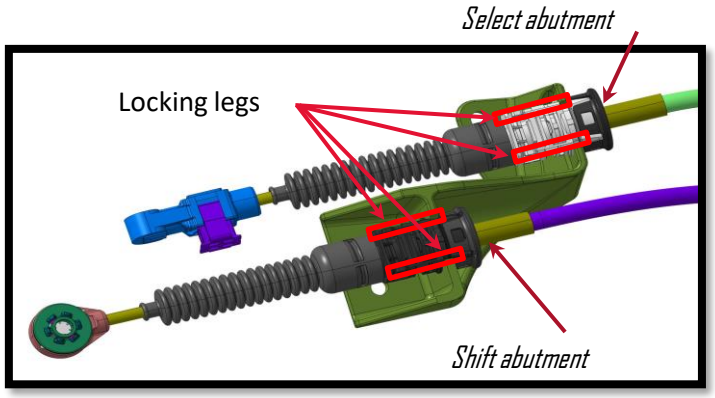
NO QUICK WINS

Step 3. Understanding Current Situation

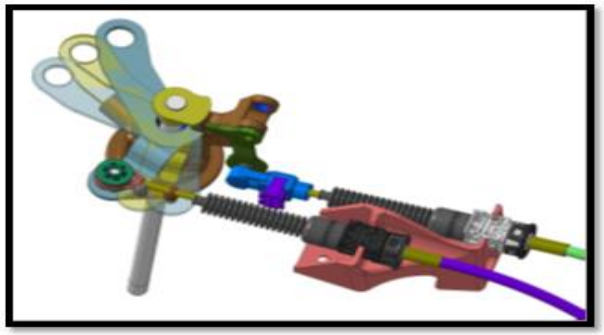


3. Understand the current situation

Fitment of abutment on chassis

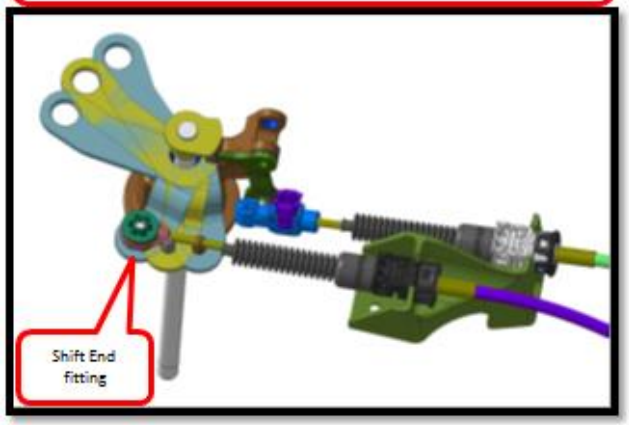


Assemble the QA body with the Select ball pin with using Special plier.



Selection of 4th gear manually
Fitment of selector cable .

Snap the Shift End fittings in to shift tower ball pins by hand



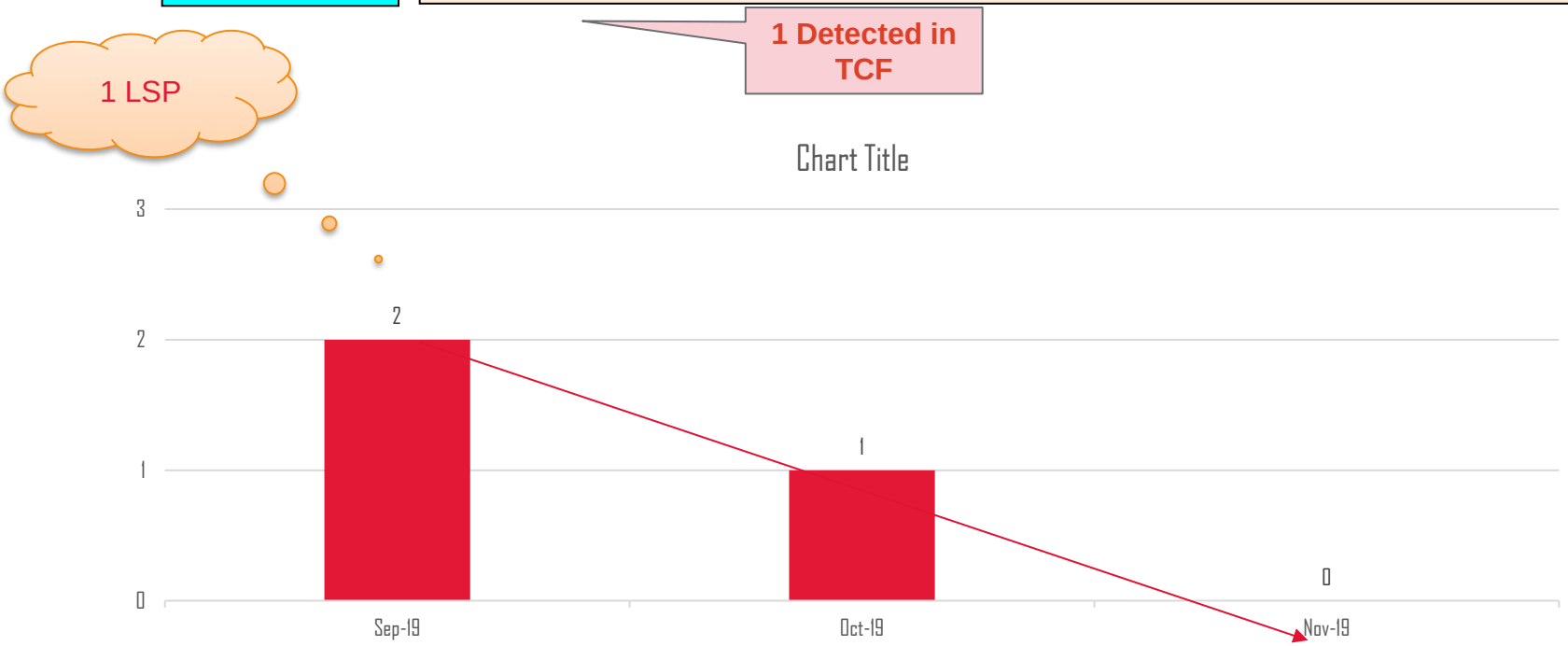
3. Understand the current situation

- Concern Reported at RFI for Reverse gear engage without operating gear selecting knob
- **All L3/ L4 Operators Working** on Stage.
- Concern Reported in Marazzo model of multi model TCF.
- Concern reported as reverse gear engage without operating gear selecting knob.
- **Rework** is done in FAI .
- **Rework** Takes more than **40 min** time.

4. Select targets

A target is a number indicating the level of improvement to be achieved.

Present Level:	Complaints reported in LSP and RFI
Target :	Concern elimination by end of Nov - 19



5. Analysis

5.1 Brain Storming

Brain Storming Team

Taj Shaikh	Mandar Joshi
Sachin Bhamre	Amit Tile
Santosh Jadhav	Pravin Patil
Satish Pagar	Nilesh Deore

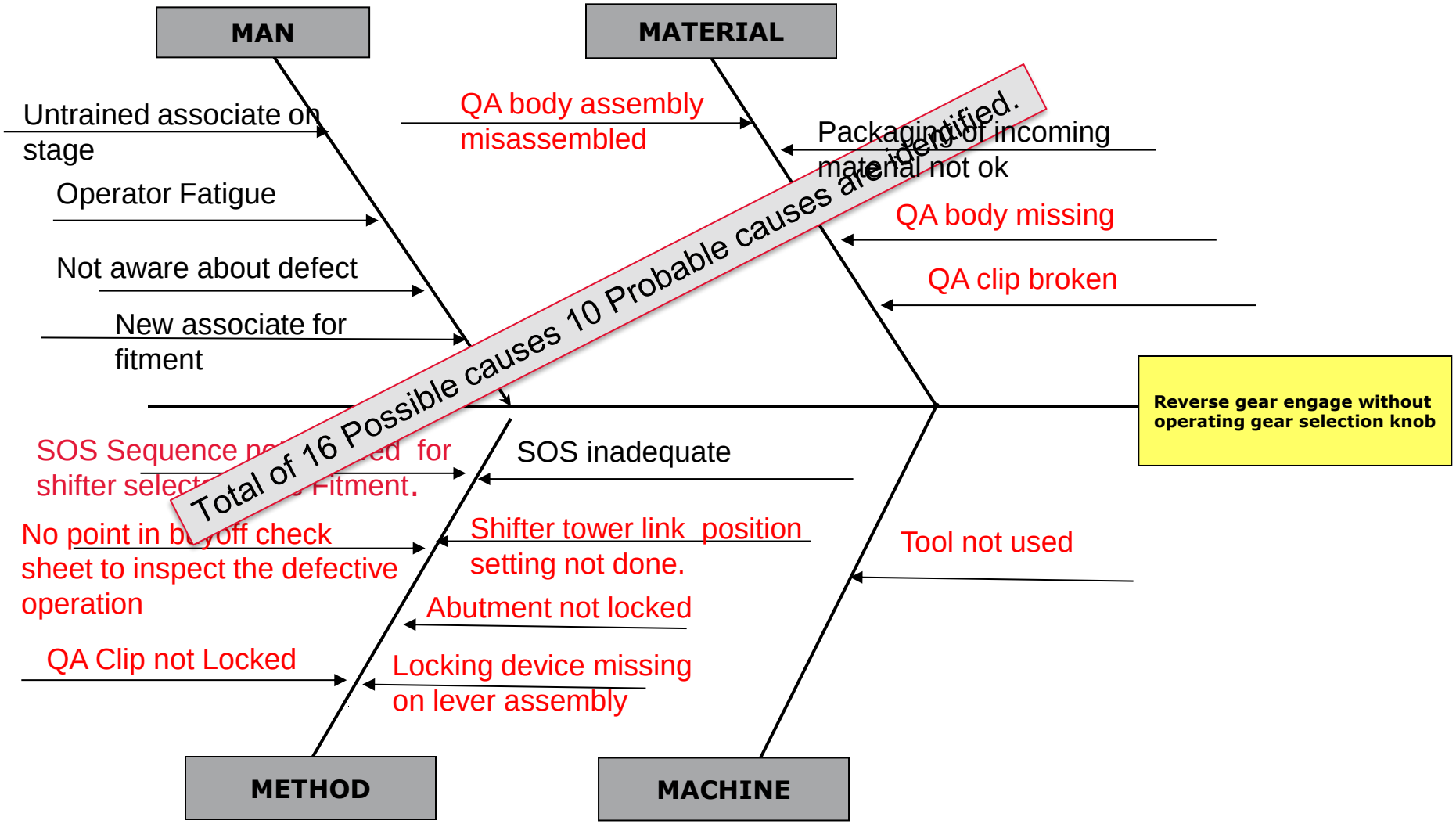
5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Probable cause
1	Untrained associate on stage
2	Operator fatigue
3	Not aware about defect
4	New associate for fitment
5	Packaging of incoming material not ok
6	QA body assembly misassembled
7	QA body missing
8	QA clip missing
9	SOS sequence not followed for shifter and selector cable fitment
10	No point in buyoff check sheet to inspect the defect operation
11	QA clip not Lock.
12	SOS inadequate .
13	Shifter tower link position setting not done .
14	Abutment not locked .
15	Locking device missing on lever assembly .
16	Tool not used .

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr no	Possible Causes	Taj shaikh	Santosh jadhav	Satish Pagar	Mandar Joshi	Amit tile	Nilesh deore	Chandu Jadhav	Total
1	Untrained associate on stage .	3	3	3	3	3	3	3	21
2	operator fatigue.	1	1	1	1	1	1	1	7
3	Not aware about defect.	3	3	3	3	3	3	3	21
4	New associate for fitment	3	3	3	3	3	3	3	21
5	Packaging of incoming material not ok	3	3	3	3	3	3	3	21
6	QA body assembly misassembled	3	3	1	3	3	1	3	17
7	QA body missing	1	3	1	3	3	1	3	15
8	QA clip missing	3	3	3	3	3	3	3	21
9	SOS sequence not followed for shifter and selector cable fitment	9	9	3	9	9	9	9	57
10	No point in buyoff check sheet to inspect the defect operation	9	3	9	3	9	3	9	45
11	QA clip not lock .	9	9	9	9	9	9	9	63
12	SOS inadequate	3	3	3	3	3	3	3	21
13	Shifter tower link position setting not done	9	9	9	9	9	9	9	63
14	Abutment not locked	9	3	9	3	9	3	9	45
15	Locking devide missing on lever assembly	9	3	9	3	9	9	9	51
16	Tool not used	9	9	9	9	9	9	9	63

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 45.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr no	Possible Causes	Reasons
1	Untrained associate on stage .	All associate found trained
2	operator fatigue.	No operator fatige observe
3	Not aware about defect.	Associate are aware about defect
4	New associate for fitment	All tranined associate for fitment
5	Packaging of incoming material not ok	Packaging found as per standard
6	QA body assembly misassembled	Part found ok
7	QA body missing	All part check and found ok
8	QA clip missing	Lock provided on selector cable for QA clip missing found ok
12	SOS inadequate	SOS found adequate

5.1 Identify Probable cause

Sr nos	List of Probable cause
9	SOS sequence not followed for shifter and selector cable fitment
10	No point in buyoff check sheet to inspect the defect operation
11	QA clip not lock .
13	Shifter tower link position setting not done
14	Abutment not locked
15	Locking devide missing on lever assembly
16	Tool not used

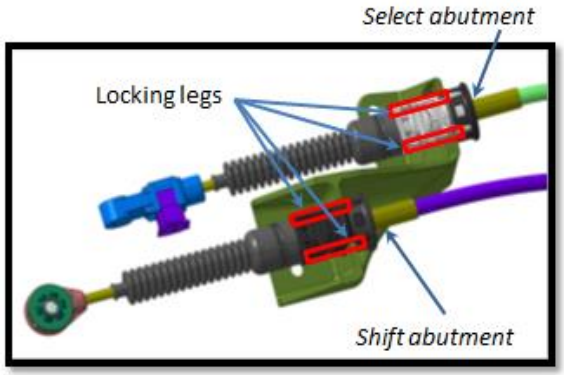
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	SOS not followed for shifter ,selector fitment	Visual inspection	During observation it is found that SOS followed by associate	**Hypothesis Invalid**

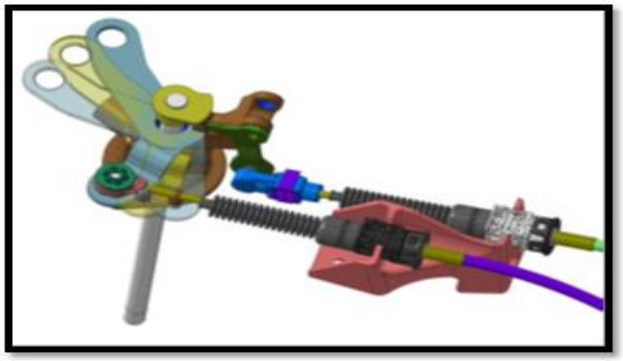
Assembly of abutment to the TM bracket

1



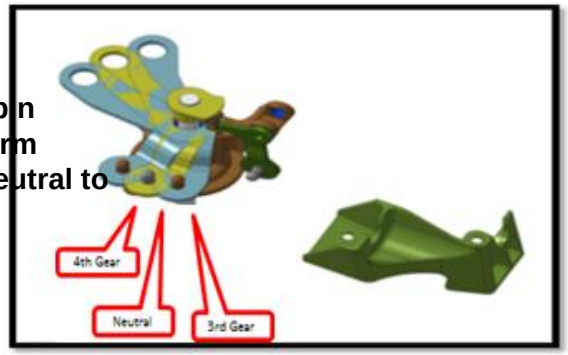
Fitment of Selector cable on ball pin and QA clip fitment

2



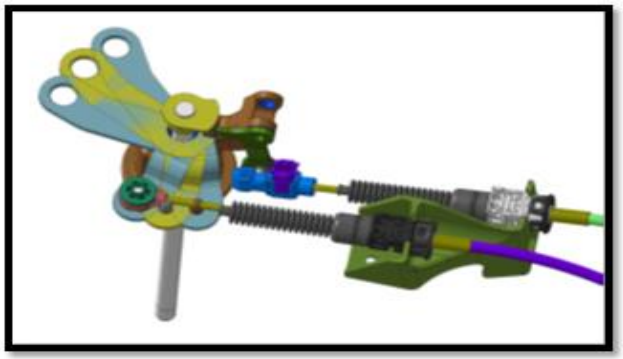
shift tower ball pin move the shift arm position from Neutral to 4th Gear

3



Shift End fittings in to shift tower ball pins by hand

4

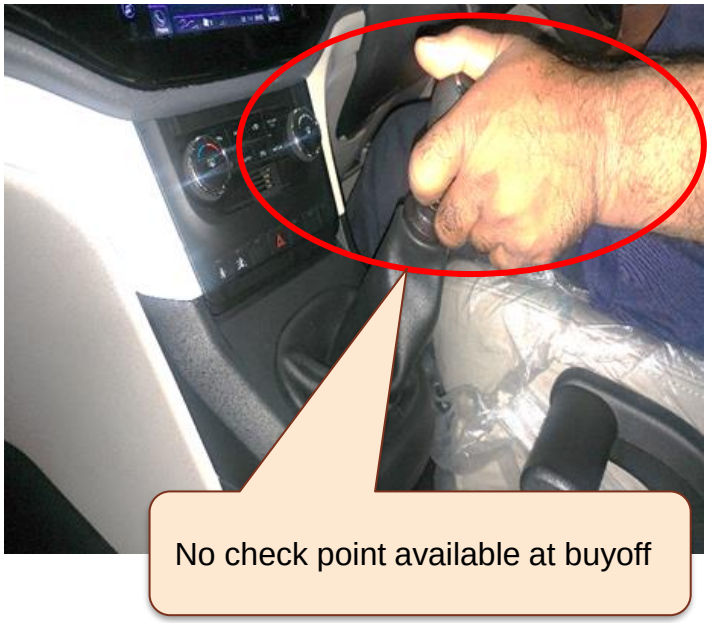


Inference:It is observe that SOS followed by associate

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
10	No point in buyoff check sheet to inspect the defective operation.	Reverse gear engage without up liftment of gear	Checked for 20 number of vehicles, no any check point available at roll down buyoff	Hypothesis valid

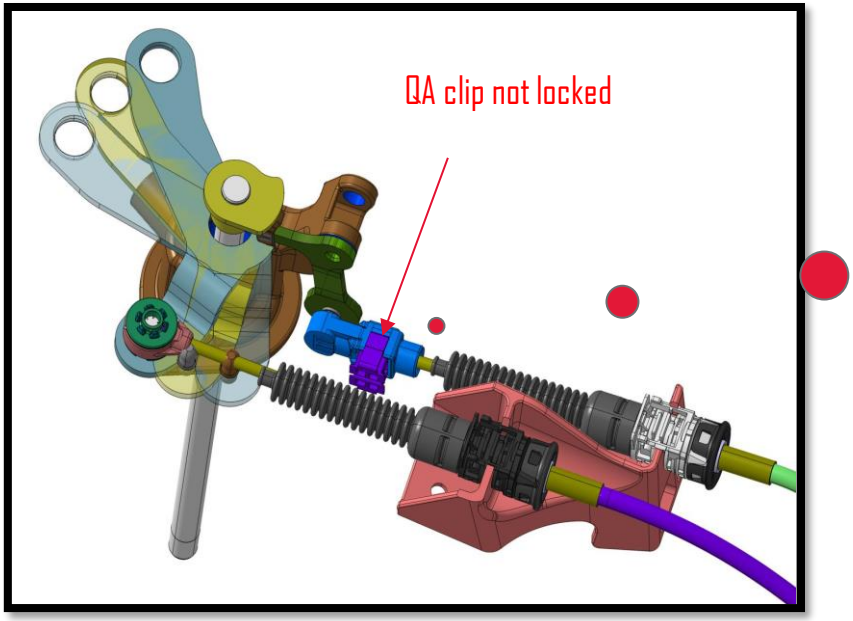


Concern	No check point available for Reverse gear engage without upliftment of knob .	Check point at buyoff
Vehicle Checked		No check point

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
11	QA clip not lock	Visual inspection	During observation it is found that QA clip may not get fitted (pressed) by associate during selector cable fitment	Hypothesis valid

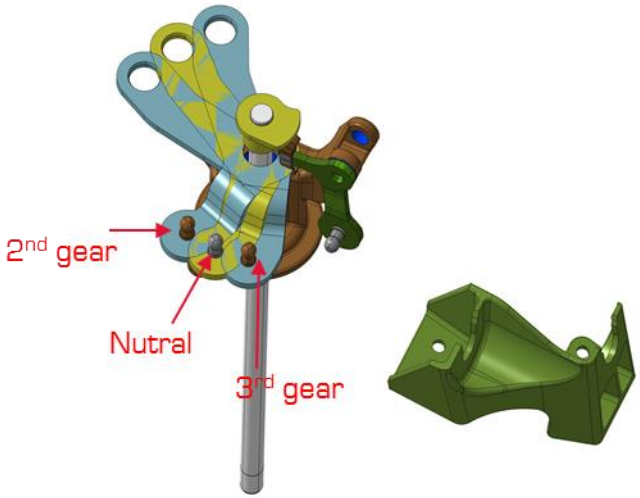


QA clip not locked it may create gear shifting problem during gear changing.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
13	Shifter tower link position setting not done	Visual inspection	During observation it is observe shifter cable may get fitted in 1st,2nd,3rd,5th,6th gear .Also there is not may mechanism detection for 4th gear position.	**Hypothesis valid**



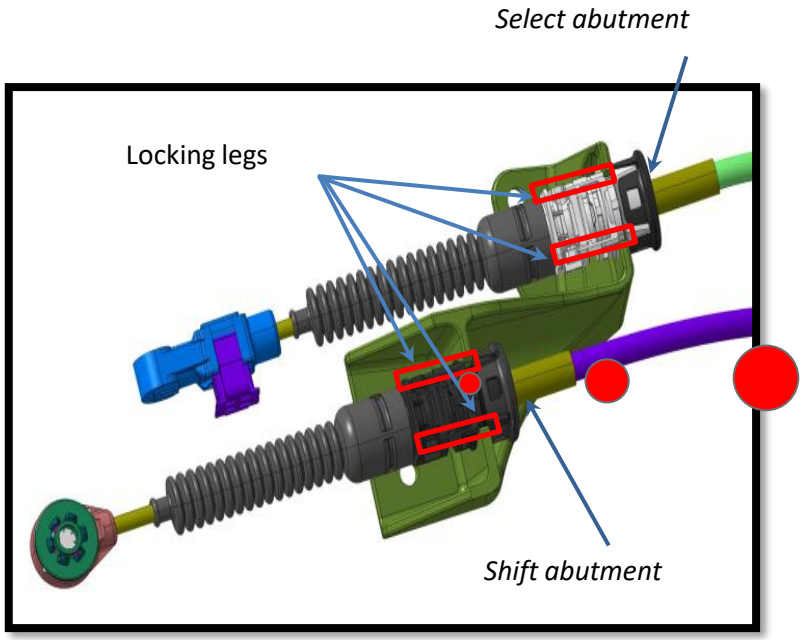
30 Vehicles checked for shifter ,selector fitment ,it is found that shifter arm can be fitted in 1st,2nd,3rd,6th gear .There is no any mechanism to recognized for identification for 4th gear position

Inference: Shifter selector fitment process was observed on gemba for Marazzo vehicles, found that associate can fit shifter cable into 1st,2nd,3rd,6th gear also .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
14	Abutment not locked	Visual inspection	During observation it is found that abutment may get poor fit, not locked .	Hypothesis valid



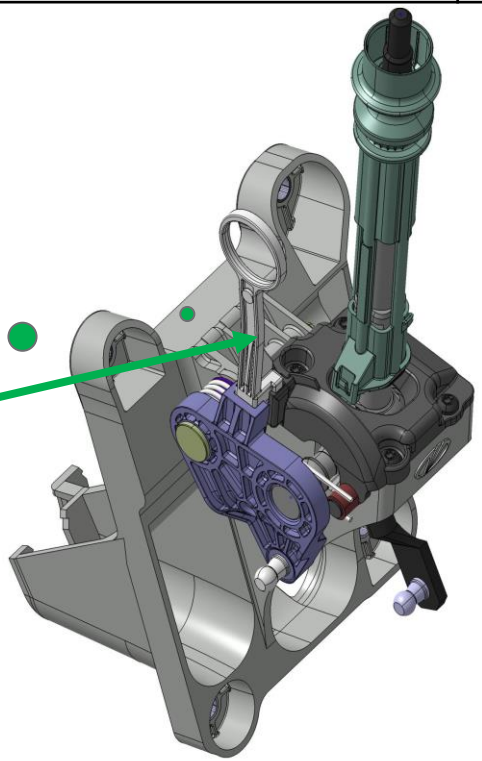
During observation it was observe that abutment may be poor fit or get skip before shifter ,selector cable fitment.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
15	Locking device missing on lever assembly .	Visual inspection	CFT member observe the stage for 2 hrs. Observations are as below… 1. Locking device checked for 25 nos of vehicle and it found ok	**Hypothesis Invalid**

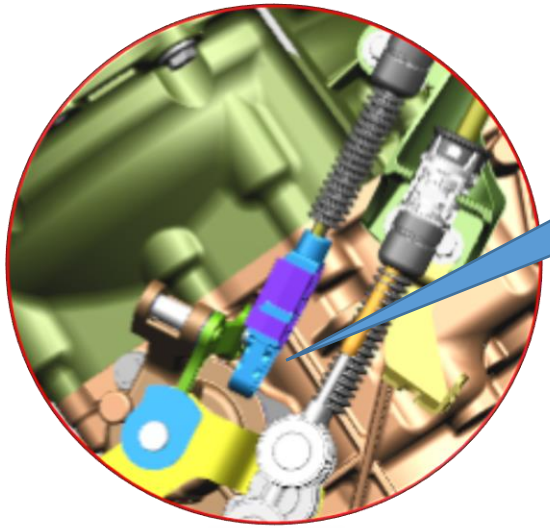
Locking device found present before shifter getting assembly on CCB



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
16	Tool not used	Visual inspection	CFT member observe the stage for 2 hrs. Observations are as below… 25 nos of vehicle check and it is observe that Tool used by associate for selector cable fitment	Hypothesis Invalid



GEAR SHIFT CABLE MOUNTED ON TRANSAXLE.

Tool used by associate for selector cable fitment .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr No	Probable Cause	Testing method	Conclusion
1	Shifter tower link position setting not done	Visual inspection	Hypothesis valid
2	No point in buyoff check sheet to inspect the defective operation.	Reverse gear engage without up liftment of gear	Hypothesis valid
3	QA clip not locked	Visual	Hypothesis valid
4	Abutment not locked	Visual inspection	Hypothesis valid

4 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Shifter tower link position setting not done	Difficult to recognized gear position .	Gear link setting order as 3rd,nutral,4th as per given spec	No alert mechanism or tool for to recognized 4th gear position .	Provide the mechanism
2	No point in buyoff check sheet to inspect the defective operation	During the check sheet finalization it gets skipped			Check point added at buyoff
3.	QA clip not locked	It is manual operation	Associate forgot to locked the QA clip	No self check available	Establish the self check
4	Abutment not locked	It is manual operation	Associate forgot to locked the QA clip	No self check available	Establish the self check

6. Implement the corrective measures

Possible Remedies Identified

Sr. No.	Root cause	Remedy Developed	Time to implement	Cost of Improve ment	Easily reversible	Complexity	Decision
1	No alert mechanism or tool for to recognized 4th gear position .	Provide the mechanism	Medium	Medium	yes	Low	**Accepted**
2	During the check sheet finalization it gets skipped	Check point added at buyoff	Medium	Medium	yes	Low	Accepted
3	No self check available	Establish the self check	Medium	Medium	yes	Low	Accepted
4	No self check available	Establish the self check	Medium	Medium	yes	Low	Accepted

6. Implement the corrective measures

Possible Remedies Identified – HEP Analysis

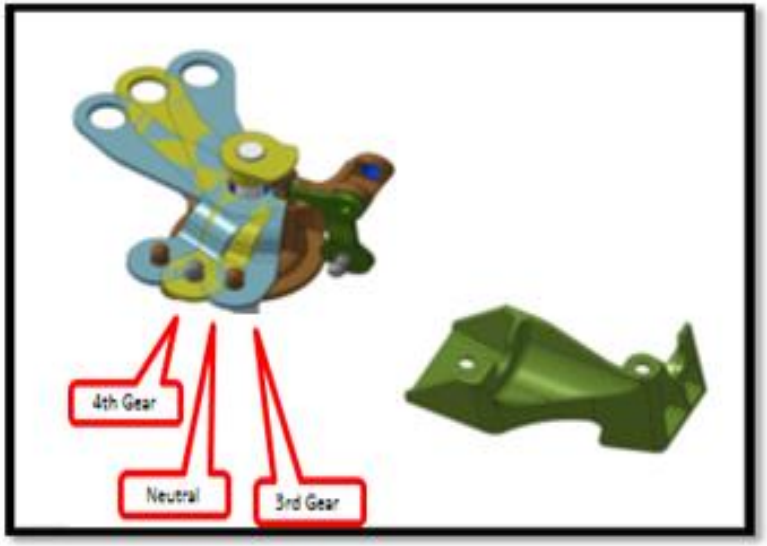
Sr.No.	List of Failure Modes and Corresponding Human errors		Applicable (Y/N)
1	Omission	-Forgetting to take parts, to read buildsheet, to scan barcode, use fixture, jig, guage,to clamp or mount fixture, use oil, promotor, to index tool, to change bit / socket, pretightening not done, to do marking, check a checkpoint	Yes
2	Excessive/ Insufficient Repetition	-All bolts not torqued - all joints not tightened	Yes
3	Wrong Order	sequence of tightening not followed, wrong sequence of operation	Yes
4	Early/Late Execution	- disconnecting pokayoke before test completion, working before or beyond pitch (stage) assigned	NA
5	Execution of Restricted Work	- working beyond pitch (stage) assigned	Yes
6	Incorrect Selection	-Selecting the wrong parts s per model variant , as per interior, wrong hardwares -Using wrong tools or wrong templates - Using wrong fixtre, guage	Yes
7	Incorrect Counting	- Carrying excessive parts - subassembly inventory more or less than assigned numbers	NA
8	Misrecognitio n	-Misreading buildsheets - Incorrect readings from instruments -Failing to determine defects, dents, damages	Yes
9	Failing to Sense Danger	-Failing to sense hot parts, rotating or sharp edges -Failing to sense areas not to be approached - pits, platforms	NA
10	Incorrect Holding	-Holding tools, guages, - wrong holding of parts not supporting parts correctly during subassembly or during fitment	Yes
11	Incorrect Positioning	-Fitment of parts in the wrong position, plugging in incorrect holes, -Transporting materials to the wrong place - Feeding materials to wrong bin	Yes
12	Incorrect Orientation	-Fitting parts inverted, wrong alignment - wrong orientation of tools, guages, fixtures	NO
13	Incorrect Motion	-Incorrect routing of wiring harness, cables - Incorrect switch setting	NA
14	Improper Holding	-Incorrect holding of parts, tools during fitment -Incorrect holding of parts, tools in fixtures -Dropping parts, products, or tools	NA
15	Inaccurate Motion	-Setting materials/parts in accurate positions -Tightening bolts to inaccurate torques	Yes
16	Insufficient Avoidan ce	- Knocking parts/tools into each other - Falling/stumbling -Touching machine switches without intention - placing parts in vehicle vulnerable to scratches or damages	NA

Note - Inputs / proposed actions related to applicable fields to be written in remarks column of FTR checksheet.

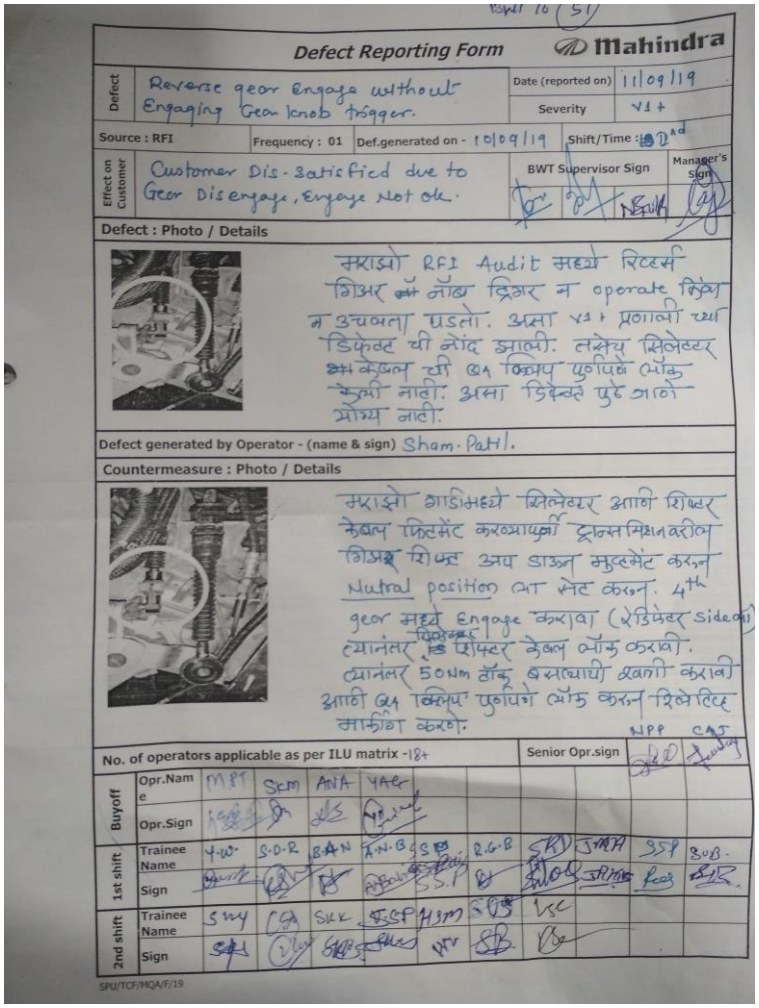
Analysis of Current Process for Reverse gear getting engaged without operating gear selection knob.–
Out of 16 types of errors 9 Errors can be occurred in this process

6. Implement the corrective measures

No	Root Cause	Action plan	Actual date	Responsibility	Status
1.	Shifter tower link position setting not done . No identification or alert mechanism for selecting in 4th gear .	Gauge/ fixture provided such that gauge will only enters into the shift arm slot when it is in 4th gear position means gauge can not enters into the slot when shift arm is in 1st, 2nd, 3rd, 5th, 6th, neutral & reverse postions. so associate can easily identified the 4th gear position.	11.9.19	Taj Shaikh	CLOSED



6. Implement the corrective measures



Defect Reporting Form — Mahindra

Defect	Reverse gear Engage without Engaging Gear knob trigger.	Date (reported on)	11/09/19
		Severity	V1+
Source : RFI	Frequency : 01	Def.generated on - 10/09/19	Shift/Time : 2nd
Effect on Customer	Customer Dis-satisfied due to Gear Disengage, Engage Not ok.	BWT Supervisor Sign	Manager's Sign

Defect : Photo / Details

मराझो RFI Audit मध्ये रिव्हर्स गिअर नॉब ट्रिगर न operate किंवा न उचलता पडतो. असा V1+ प्रकारची ही डिफेक्ट ची नोंद झाली. तसेच सिलेक्टर केबल ची QA क्लिप पुर्णपणे लॉक केली नाही. असा डिफेक्ट पुढे जाणे योग्य नाही.

Defect generated by Operator - (name & sign) Sham. Patil.

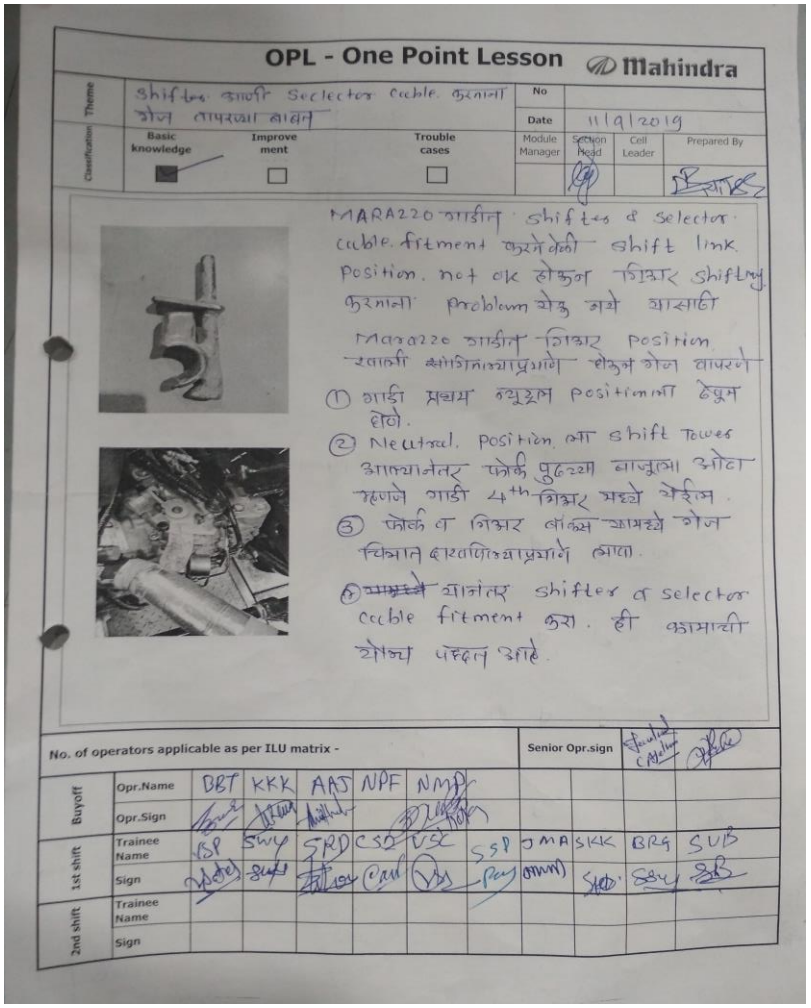
Countermeasure : Photo / Details

मराझो गाडीमध्ये सिलेक्टर आणि शिफ्टर केबल फिटमेंट करण्यापूर्वी ट्रान्समिशन वरील गिअर शिफ्ट अप डाऊन मुव्हमेंट करून Nutral position ला सेट करून. 4th gear मध्ये Engage करावा (रेडियेटर side ला) त्यानंतर शिफ्टर केबल लॉक करावी. त्यानंतर 50Nm टॉर्क बसवाची खात्री करावी आणि QA क्लिप पुर्णपणे लॉक करून रिलेटिव्ह मार्किंग करणे.

No. of operators applicable as per ILU matrix -18+ | Senior Opr.sign | NPP | CAJ

Buyoff	Opr.Name	MBT	SKM	ANA	YAC							
	Opr.Sign											
1st shift	Trainee Name	Y.W.	S.D.R	B.A.N	K.N.B	S.S.P	R.G.B	SKD	JMA	SSP	S.D.B.	
	Sign											
2nd shift	Trainee Name	SWY	CGA	SKK	SSF	HJM	SVB	VSC				
	Sign											

SPU/TCF/MQA/F/19



OPL - One Point Lesson — Mahindra

Theme	Shifter आणि Selector cable करताना गेअर वापरणे बाबत			No			
				Date	11/9/2019		
Classification	Basic knowledge ■	Improvement ☐	Trouble cases ☐	Module Manager	Section Head	Cell Leader	Prepared By

MARA220 गाडीत shifter & selector cable fitment करणेकडे shift link position. not ok होऊन गिअर shifting करताना problem येतो म्हणून

Mara220 गाडीत गिअर position खालील सोयीनुसार होऊन गेअर वापरणे

① गाडी प्रत्यक्ष न्यूट्रल position ला ठेवून होते.

② Neutral position ला shift Tower आत्यानंतर फोर्क पुढच्या बाजुला ओढा म्हणजे गाडी 4th गिअर मध्ये येईल.

③ फोर्क व गिअर बॉक्स मध्ये गेअर चिन्हाप्रमाणे लावा.

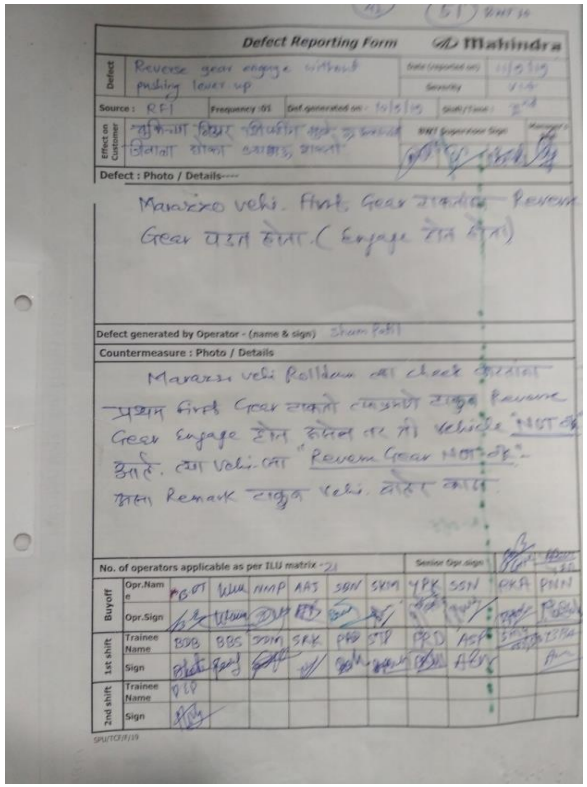
④ त्यानंतर shifter & selector cable fitment करा. ही कामाची योग्य पद्धत आहे.

No. of operators applicable as per ILU matrix - | Senior Opr.sign | CAJ | Mathew

Buyoff	Opr.Name	BBT	KKK	AAJ	NPF	NMP						
	Opr.Sign											
1st shift	Trainee Name	VSP	SWY	SRD	CSD	VSC	SSP	JMA	SKK	BRG	SVB	
	Sign											
2nd shift	Trainee Name											
	Sign											

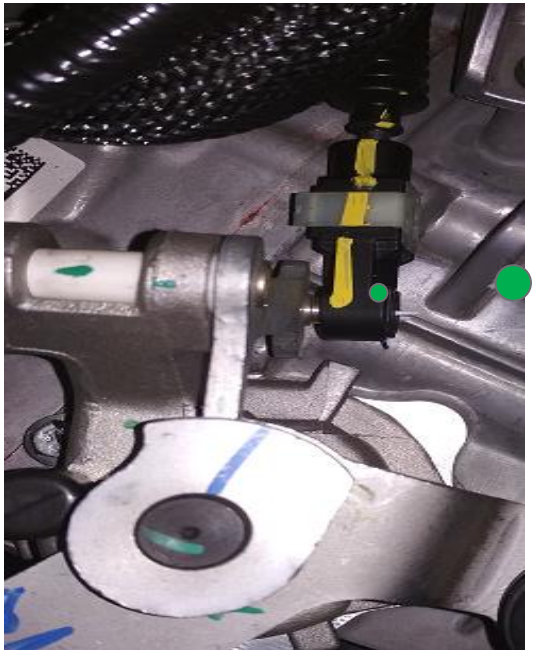
6. Implement the corrective measures

No	Root Cause	Action plan	Actual date	Responsibility	Status
2	No point in buyoff check sheet to inspect the defective operation	1) Check point added at buyoff check sheet. 2) Communicate done through OPL	11.9.19	Taj Shaikh	Close



6. Implement the corrective measures

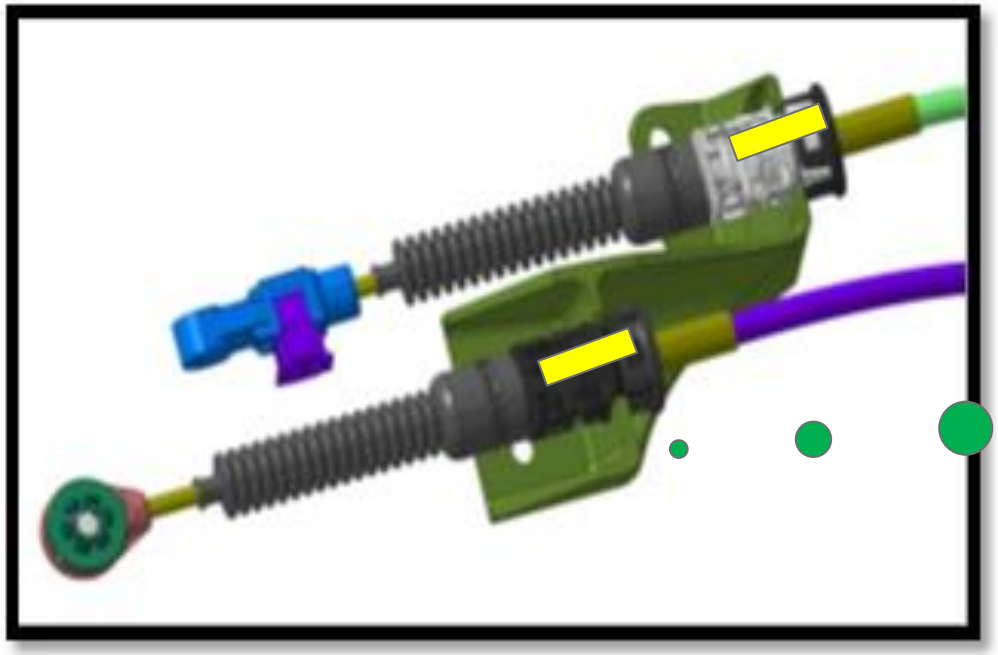
No	Root Cause	Action plan	Actual date	Responsibility	Status
3	QA Clip not lock	Match marking started on QA clip with yellow paint	11.9.19	Taj Shaikh	CLOSED



Match marking on QA clip with yellow marker

6. Implement the corrective measures

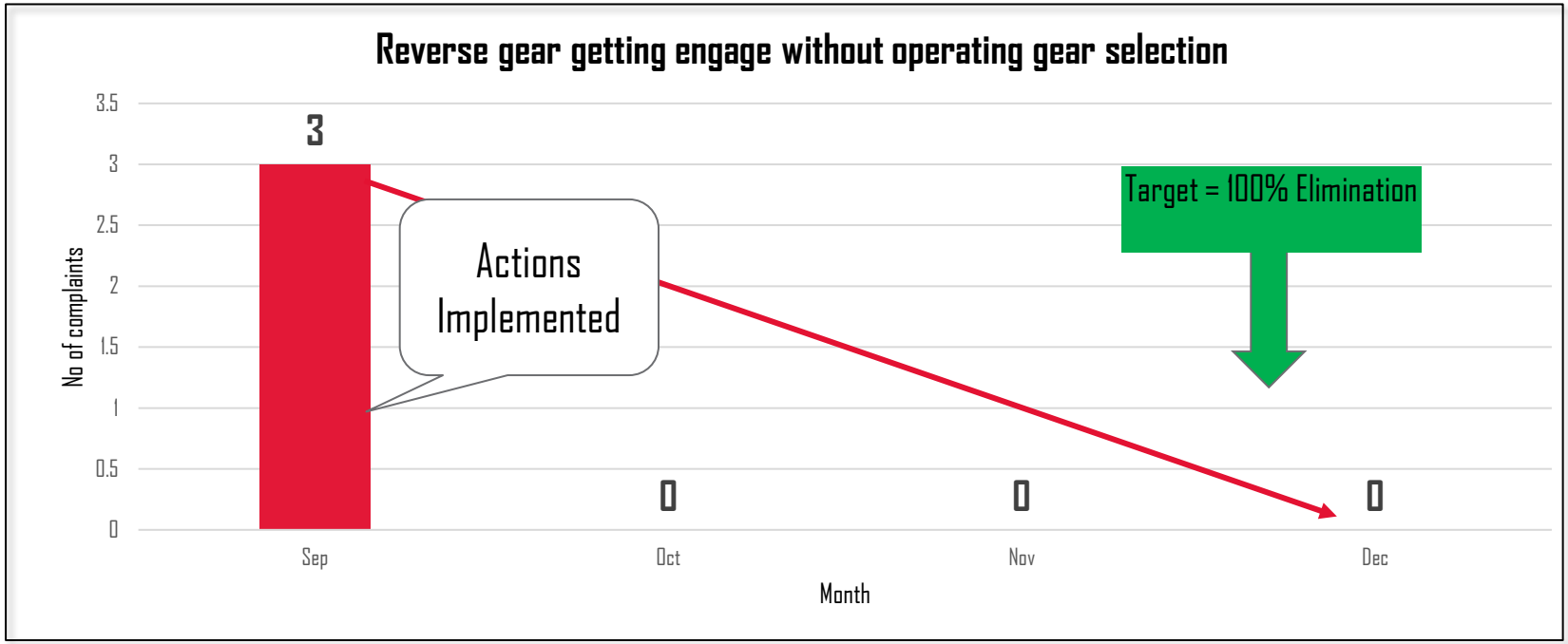
No	Root Cause	Action plan	Actual date	Responsibility	Status
4	Abutment not lock	Match marking started on abutment with yellow paint	11.9.19	Taj Shaikh	CLOSED



Match marking on Abutment with yellow marker

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend



Inference – After Implementing Scan Based mechanism Pokayoke for torqueing concern and traceability for the same monitored for 4 months No issue reported at Internal Audit and well as Field.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	**Actual date**	Responsibility	Status
1	Gauge/ fixture provided such that gauge will only enters into the shift arm slot when it is in 4th gear position means gauge can not enters into the slot when shift arm is in 1st, 2nd, 3rd, 5th, 6th, neutral & reverse postions. so associate can easily identified the 4th gear position.	11.9.19	Taj Shaikh	CLOSED
2	No point in buyoff check sheet to inspect the defective operation.	11.9.19	Taj Shaikh	CLOSED
3	QA clip match marking started	11.9.19	Taj Shaikh	CLOSED
4.	Abutment match marking started	11.9.19	Taj Shaikh	CLOSED
3	SOS Modification	11.9.19	Nilesh Deore	CLOSED
4	PFMEA updation	6.11.19	Suhas Aher	CLOSED
5	PFC updation	6.11.19	Pravin Patil	CLOSED
6	QA matrix	11.9.19	Nilesh deore	CLOSED

Check & confirm the results & effects

7.3 Tangible & Intangible benefits:

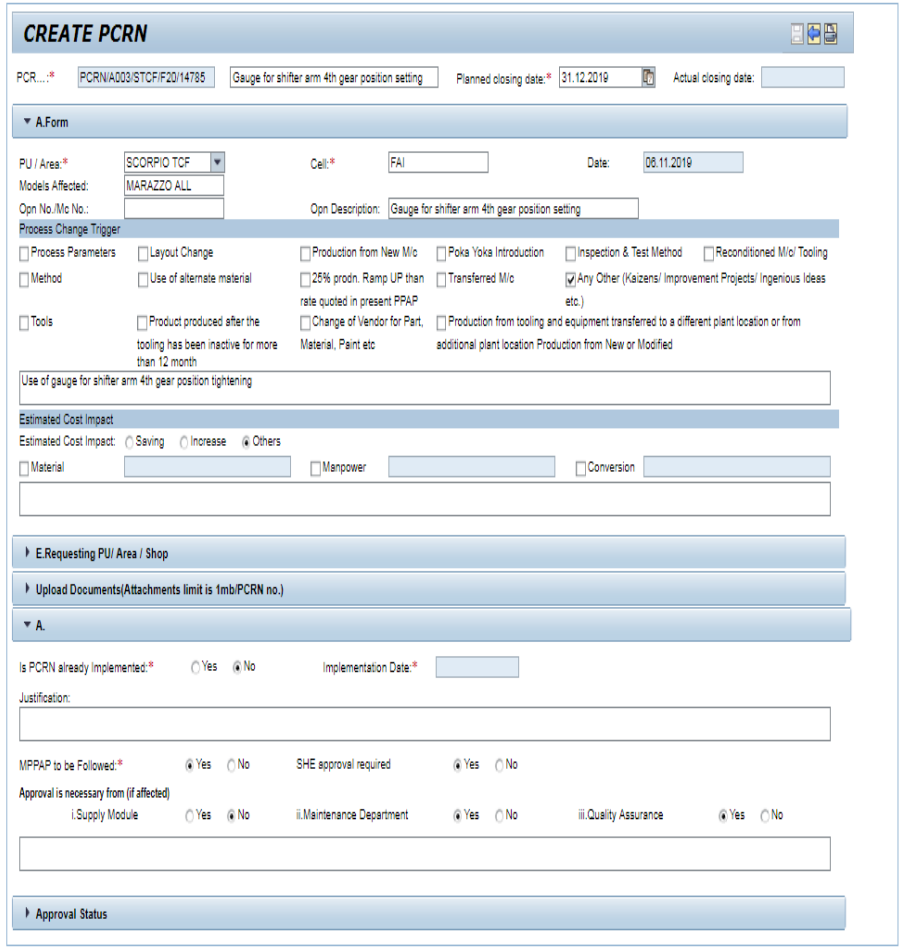
Tangible Benefit:-

- One Z stage or V1+ level operation stage error proofing done.
- Regeneration of Field complaint eliminated.
- All the possible sources of failures are eliminated from stage.

Intangible Benefit:-

- Associate complexity eliminated
- Associate Moral Improved.

Process Changed Request Note – Approved by all stockholders.



CREATE PCRN

PCR...:* PCRN/A003/STCF/F20/14785 | Gauge for shifter arm 4th gear position setting | Planned closing date:* 31.12.2019 | Actual closing date:

A.Form

PU / Area:* SCORPIO TCF | Cell:* FAI | Date: 08.11.2019

Models Affected: MARAZZO ALL

Opn No./Mc No.: | Opn Description: Gauge for shifter arm 4th gear position setting

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part, Material, Paint etc
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

Use of gauge for shifter arm 4th gear position tightening

Estimated Cost Impact

Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

- [ ] Material
- [ ] Manpower
- [ ] Conversion

E.Requesting PU/ Area / Shop

Upload Documents(Attachments limit is 1mb/PCRN no.)

A.

Is PCRN already Implemented:* ○ Yes ◉ No | Implementation Date:*

Justification:

MPPAP to be Followed:* ◉ Yes ○ No | SHE approval required ◉ Yes ○ No

Approval is necessary from (if affected)

i.Supply Module ○ Yes ◉ No | ii.Maintenance Department ◉ Yes ○ No | iii.Quality Assurance ◉ Yes ○ No

Approval Status

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	PCRN Implemented PCRN/A003/STCF/F19/14785	6.11.19	Sayali More	Closed
2.	SOS Modification.	11.9.19	Nilesh deore	Closed
3.	PFMEA updation.	6.11.19	Suhas Aher	Closed
4	PFC updation	6.11.19	Pravin Patil	Closed

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	On job Training given to operator.	11.9.19	Nilesh Deore	Closed

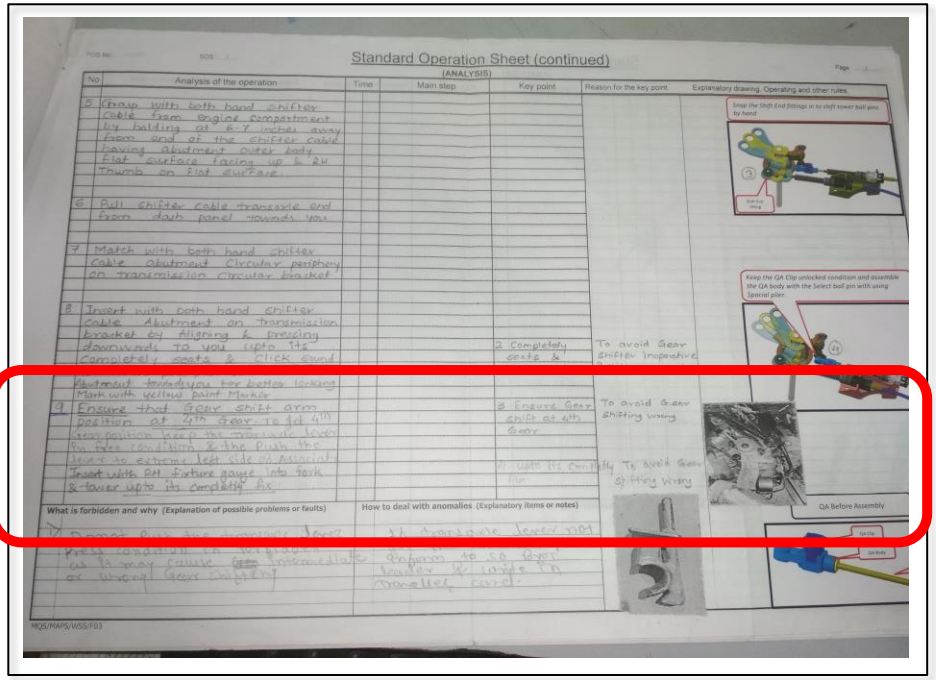
8.3 Additional Audit & check

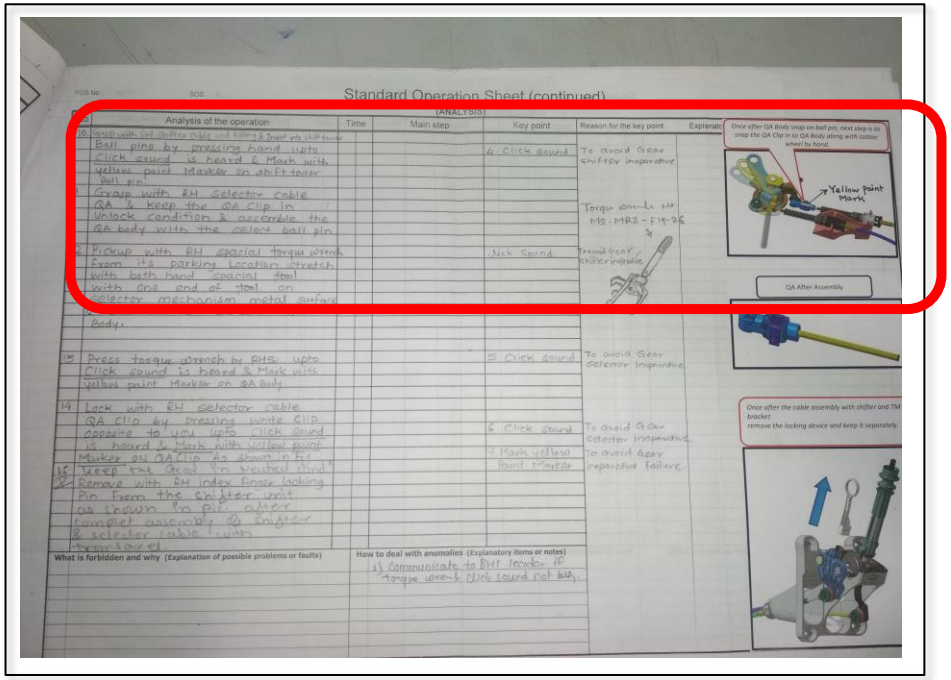
1	Check point at Rolldown buyoff	11.9.19	Taj Shaikh	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SOS Updation





SOS updated for Reverse gear getting engage without operating gear selection knob

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

PCRN implemented PCRN/A003/STCF/F19/14785

CREATE PCRN

PCR...:* PCRN/A003/STCF/F20/14785 Gauge for shifter arm 4th gear position setting Planned closing date:* 31.12.2019 Actual closing date:

A.Form

PU / Area:* SCORPIO TCF Cell:* FAI Date: 06.11.2019
Models Affected: MARAZZO ALL
Opn No./Mc No.: Opn Description: Gauge for shifter arm 4th gear position setting

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part, Material, Paint etc
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

Use of gauge for shifter arm 4th gear position tightening

Estimated Cost Impact

Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

- [ ] Material
- [ ] Manpower
- [ ] Conversion

E.Requesting PU/ Area / Shop

Upload Documents(Attachments limit is 1mb/PCRN no.)

A.

Is PCRN already Implemented:* ○ Yes ◉ No Implementation Date:*

Justification:

MPPAP to be Followed:* ◉ Yes ○ No SHE approval required ◉ Yes ○ No

Approval is necessary from (if affected)

i.Supply Module ○ Yes ◉ No ii.Maintenance Department ◉ Yes ○ No iii.Quality Assurance ◉ Yes ○ No

Approval Status

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

FMEA Updation

Plant : Nashik			Model :Marazzo												Cross Functional Team				
Part No/s.:			Part Name/ Process:																
Latest Change Letter:			Optn No.:0703007												Vikrant Ghule	Santosh Jadhav			
Plant Shop :Scorpio TCF			Optn Desptn: SHIFTER AND SELECTOR CABLE CONNECTION TO TRANSAXLE												Sayali More				
Module/ Cell :Final			M/C Name & No.:																
Process Function	**Requirements**	**Potential Failure Mode**	**Potential Effect(s) of Failures**	**sev**	**class**	**Potential Cause(s)/Mechanism(s) of Failures**	**Current Process Controls Prevention**	**Occur**	**Current Process Controls Detection**	**Detect**	**RPN**	**Recommended Action(s)**	**Responsibility & Target Completion Date**	**Action Results**					
														Action Taken	**Sev**	**Occ**	**Det**	**RPN**	
CUTTING OF STRAP OF SHIFTER/ SELECTOR CABLE	NO DAMAGE OR CUT MARK TO ADJUSANT RUBBER, EGR PIPE, FUEL LINE	RUBBER HOSE, EGR PIPE, FUEL PIPE CUT **(FIELD COMPLAINT DEC 18)**	**Mfg. & Assy:** REWORK **End user:** VEHICLE INOPRABLE	8		ASSOCIATE PUT CUTTUNG PLIER IN ENGIBE COMPARTMENT OF RUBBER HOSE , EGR PIPE, FUEL LINE DURING HIS WORK	OPERATOR TRAINING	5	SELF CHECK	4	160	1) Part such as bin,gun,piler,cutter kept on engine compartment eliminate from process 2) Position of vin plate change 3) Senior associate and superviser process audit checksheet updated 4) Checkpoint added at console stage compartment checking	NILESH/ AJIT DEC 18	ACTION COMPLETED					
CONNECT THE SHIFTER [A]& SELECTOR [B] CABLE INTO TRANSMISSION BRACKET [C] AS SHOWN IN FIGURE	SELECTOR AND SHIFTER CABLE CONNECTION IN TRANSMISSION	IMPROPER LOCKING	**Mfg. & Assy:** REWORK **End user:** Nil	3		OPERATOR MISTAKE	OPERATOR TRAINING	5	VISUAL INSPECTION/ CLICK SOUND DURING CONNECTION	4	60								
CONNECT THE SHIFTER [A]& SELECTOR [B] CABLE INTO GEAR SHIFT MECHANISM [D] AND LOCK THE CLIP [P] AS SHOWN IN FIGURE	SELECTOR AND SHIFTER CABLE CONNECTION IN GEAR SHIFT MECHANISM.	IMPROPER LOCKING **(FIELD COMPLAINT)**	**Mfg. & Assy:** REWORK **End user:** Nil	3		OPERATOR MISTAKE	OPERATOR TRAINING	5	VISUAL INSPECTION/ CLICK SOUND DURING CONNECTION	4	60	1. SPECIAL TOOL TO BE MADE FOR FULLY LOCKING OF SHIFTER/SELECTOR CABLE CONNECTION ON GEAR LEVER MECHANISM	PRAVIN PATIL	ACTION COMPLETED 20.02.2019					
	VEHICLE IN 4 TH GEAR WHILE DOING CONNECTION ON GEAR SHIFT MECHANISM	OPERATOR SELECTS WRONG POSITION OF GEARS OTHER THAN 4TH	**Mfg. & Assy:** GEAR SHIFTING WILL NOT HAPPEN **End user:** Nil	3		OPERATOR MISTAKE	PROCESS MENTION IN SOS	5	GERA SHIFTING AT CHECK AT 1. CONSOLE STAGE 2.ROLL DOWN STAGE 3.CAI	7	105	Gauge to be use to put gear arm in fourth gear position.	PRAVIN PATIL	Action completed 06.11.2019					
	QA CLIP FULLY LOCK	IMPROPER LOCKING	**Mfg. & Assy:** GEAR SHIFTING WILL NOT HAPPEN **End user:** Nil	3		OPERATOR MISTAKE	PROCESS MENTION IN SOS	5	GEAR SHIFTING CHECK AT 1. CONSOLE STAGE 2.ROLL DOWN STAGE 3.CAI	7	105								

Mahindra & Mahindra Ltd.

Process Potenial Failure Mode & Effect Analysis.

Rev. No : 08 | Doc No.: MQS/PFMEA/NASHIK/U321/TCF/0703007

Reviewed by :Suhas Aher

Rev. Date : 06.11.2019 | FMEA Date (Orig) :18-07-2017

Approved by : Mandar Joshi

Rev No.	*Rev Date*	*Revision Summary*	*Reason for Revision*
01	18-07-2017	New Release	
02	21.02.2018	Change in OCC	Six Monthly Review
03	07.08.2018	NO Change	Six Monthly Review
04	06.12.2018	Fuel overflow pipe cut	FIELD COMPLAINT
05	12.02.2019	Change in OCC	Six Monthly Review
06	20.02.2019	Splecial tool implementation	FIELD COMPLAINT
07	05.08.2019	NO Change	Six Monthly Review
08	06.11.2019	PCRN - Introduction of new gauge for shifter arm in 4th gear postion setting	PCRN/A003/STCF/20/14785

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

QA Matrix Updation

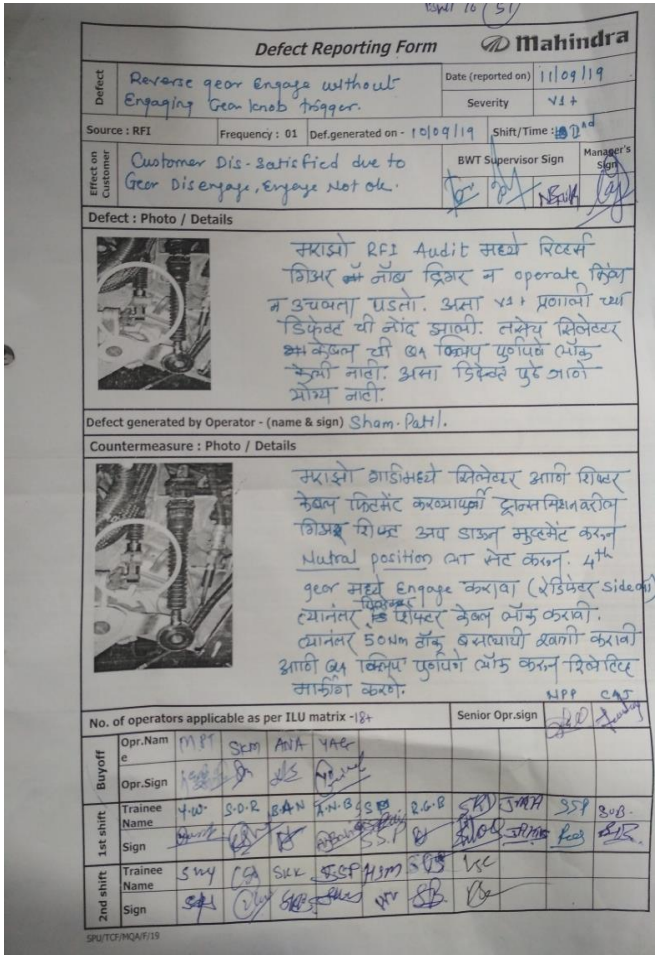
DEPT NAME	SXTCF	SIGNATURES					
SECTION NAME	FAI	Dept Head	Rahul Narkar			Section	0.61
		Module Manager	Taj Shaikh			Plant	100
		Section Manager	1	2	3	4	Revision date	12.09.19

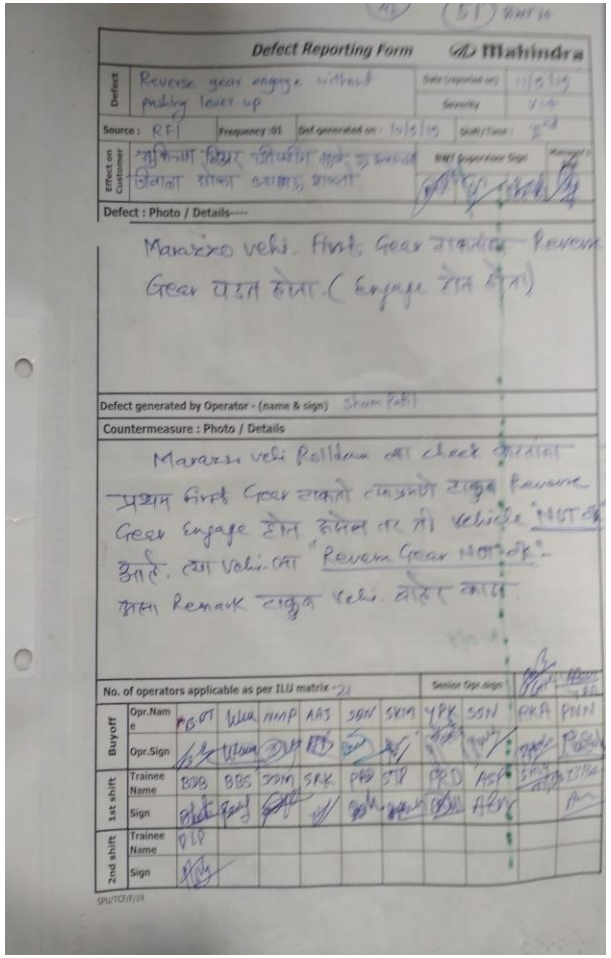
| Workstation No. | Process description (POS) | Date | Failure modes | Severity | Prev. occurence | Potential defect | Inspection process: Section | | | | | | |

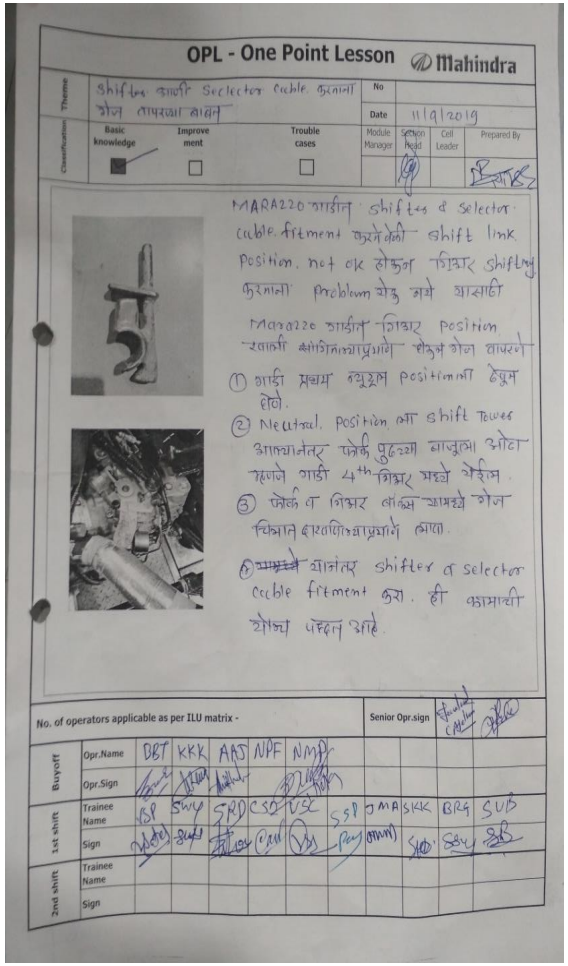
8. Standardization

8.2 Training:

Operator communication and On job Training given to operators : Resp Taj : Sep 19 : Closed







9. Summaries & Plan future actions

9.1 Conclusion:

Elimination of reverse gear getting engaged without operating gear selection knob .

9.2 Horizontal Deployment:

System given for implementation at Nashik Bolero Line, Haridwar, Zahirabad & Chakan plants

9.3 Lessons Learnt:

Things Gone Wrong : Associate complexity not consider with increasing number of Models on same line.

Things Gone Right : Using the HEP principles we are able to eliminate the all failure modes & able to make the operation error proof.

Thank you

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